

Product Requirements Document

Ufaq Khalid — Personal Portfolio Website

Complete PRD for the shipped portfolio: goals, users, routes, features, content, motion, acceptance criteria, and known gaps. Design tokens & animation detail also live in the Design Sheet PDF.

Field · Value

****Product name**** · Ufaq Portfolio (`ufaq-portfolio`)

****Owner**** · Ufaq Khalid

****Version**** · 1.0 (reflects shipped product)

****Status**** · Shipped / under academic code review

****Primary deploy**** · Vercel (GitHub repo `my-portfolio`)

****Live URL**** · <https://my-portfolio-ufaq-khalid.vercel.app>

****Document type**** · PRD (requirements + acceptance) — separate from Design Sheet

1. Executive summary

Build and maintain a personal portfolio website that presents Ufaq Khalid as a Software Engineer / Full-Stack Builder. The site must convince professors (viva / coursework review) and employers that coursework and original builds are clear, usable, and professionally presented — with case studies, skills, services, and a working contact path.

This PRD describes what the product must do, for whom, with what acceptance criteria. Visual/token details live in the Design Sheet; this document owns product requirements.

2. Problem statement

Students often show raw projects without narrative (problem !' approach !' outcome). Reviewers and recruiters need:

1. Who the person is and how they work
2. What they can build (skills / services)
3. Proof (case studies with structure, not only screenshots)
4. An easy way to contact them

Without a coherent portfolio, demos feel scattered and hard to grade or hire against.

3. Goals

3.1 Primary goals

1. Present a coherent personal brand and role in the first viewport.
2. Explain a clear build process (About).
3. Show skill groups and service offerings.
4. Ship ****three**** case studies with full narrative + architecture diagram.
5. Provide social proof (testimonials) and FAQ.
6. Enable contact via email, phone, and WhatsApp form.
7. Support ****dark**** (default) and ****light**** themes with persisted preference.
8. Deliver motion that feels premium but respects ****reduced motion****.
9. Deploy as a fast static Next.js site on Vercel.

3.2 Secondary goals

1. Desktop-only cursor trail and hero 3D as craft signals.
2. Design + motion documentation for review day.
3. Optimized project imagery (WebP / next/image).

3.3 Non-goals (out of scope)

- CMS, auth, blog, multi-language
 - Backend form/email API (WhatsApp deep-link is enough)
 - Live embedded demos / sandboxes per project
 - Analytics dashboard, A/B testing
 - E-commerce or client portal
 - Full SEO pack (OG images, sitemap) — optional future
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4. Users & personas

Persona · Goal · Must-have support

****Professor / viva examiner**** · Judge process maturity & clarity · About steps, case studies, architecture, stack, metrics

****Employer / recruiter**** · Assess fit quickly · Hero, skills, work cards, contact

****Collaborator / client**** · See services & reach out · Services, FAQ, WhatsApp/email

****Student seeking help**** · Know if mentoring is offered · FAQ answer on university project help

****Peer developer**** · Judge craft · Motion, theme, case study depth

5. User journeys

1. ****Viva reviewer:**** Land !' Hero !' Work !' open case study !' Architecture + Problem/Approach/Outcome !' Contact if needed.
 2. ****Recruiter:**** Land !' Skills !' Work !' Contact / email.
 3. ****Mobile visitor:**** Same content; no hero 3D; hamburger nav; readable cards.
 4. ****Return visitor (theme):**** Preference restored from `localStorage` without flash.
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6. Information architecture & routes

Route · Requirement

`/` · Homepage with all marketing sections in fixed order

`/projects/{slug}` · Case study detail for each project in content

`/projects/campus-admin-dbms` · Campus Admin DBMS

`/projects/dsa-arena` · DSA Arena

`/projects/ship-ready-se` · Ship-Ready SE Lab

Unknown slug · 404 (`notFound`)

Global chrome (every page): Navbar, ThemeToggle, CursorTrail (when eligible), ScrollProgress, Footer, ThemeProvider, GsapRefresh.

Home section order (required):

Hero (#home) !' About (#about) !' Skills (#skills) !' Services (#services) !' Work (#work) !' Testimonials (#testimonials) !' FAQ (#faq) !' Contact (#contact) !' Footer.

Nav links required: Home, About, Skills, Services, Work, Testimonials, FAQ. Contact reachable via CTA / #contact / footer (may omit from primary pill to reduce clutter).

7. Functional requirements

7.1 Branding & Hero

- Display name ****Ufaq Khalid**** as dominant H1 (display font).
- Role line under name (display italic).
- Supporting hero sentence from content.
- Availability status (“Available for select projects — 2026”).
- Primary CTA !' Work; secondary CTA !' Contact.
- Desktop (“e900px”): optional 3D scene or safe fallback; ****no 3D on small screens****.

7.2 About (“How I build”)

- Four numbered process steps from content.
- Desktop: scroll-linked active step + dot navigation.
- Mobile: stacked readable steps with scroll reveal.

7.3 Skills

- Four skill groups with title, description, item chips.
- Overlapping / stacked card presentation with hover emphasis.
- Keyboard focusable cards.

7.4 Services

- Four services in accordion: title, description, tags, preview image.
- Only one expanded at a time (or mutually clear open state).
- Images from local project art (not stock Unsplash).

7.5 Work (case collection)

- Exactly three project cards from content.
- Each shows category, title, summary, stack chips, “Case study” affordance.
- Click/tap navigates to `/projects/{slug}`.
- Desktop fine-pointer: 3D tilt hover; disabled for touch / reduced motion.

7.6 Case study inner pages

Each case study must include:

- Category, year, title, summary
- Role, context, year meta
- Hero image with meaningful alt text
- Metrics strip
- **Architecture** blueprint (node flowchart + data loop)
- Problem, Approach, Outcome (full paragraphs)
- Highlights list
- Stack chips
- Navigation back to `/#work`
- **No** “Ask for a demo” button (removed by requirement)

7.7 Testimonials

- At least four static quotes with name + role.
- Horizontal browse / drag on small screens.

7.8 FAQ

- At least five Q&A items including willingness to help students.
- Accordion expand/collapse with accessible expanded state.

7.9 Contact

- Show email, phone, WhatsApp as actionable links.
- Form collects name/message (and related fields as implemented) and opens WhatsApp with prefilled text.
- No server-side mail required for v1.

7.10 Footer

- Brand wordmark, CTA, copy-email, nav shortcuts, social, availability, local time (Asia/Karachi).

7.11 Theme

- Dark default; light available.
- Persist `ufaq-theme` in `localStorage`.
- Prevent theme flash with early inline script.
- Architecture blueprint may remain dark (accepted exception).

7.12 Motion & accessibility

- Honor `prefers-reduced-motion`: no trail, no tilt, no ornamental loops; content still fully readable.
- Focus-visible styles on interactive controls.
- Decorative motion must not be the only carrier of meaning.

7.13 Performance / media

- Project images served as WebP (with next/image AVIF/WebP).
- Separate small thumbs for cursor trail.
- First work card / case hero may use `priority`.

8. Content requirements

Single source of truth: src/data/content.ts.

Must define:

- `site` (name, role, contact, nav, github)
- `aboutSteps` (4)
- `skillGroups` (4)
- `services` (4)
- `projects` (3) with full case-study fields + architecture nodes
- `testimonials` (4)
- `faqs` (5)

Projects required:

1. Campus Admin DBMS — PHP/MySQL campus admin
 2. DSA Arena — Unity/C# game/algorithms
 3. Ship-Ready SE Lab — SDLC delivery pipeline
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9. Non-functional requirements

Area · Requirement

****Framework**** · Next.js App Router (v16.x), React 19
****Styling**** · Tailwind 4 + CSS variables for theme
****Motion**** · GSAP + ScrollTrigger; Framer Motion where specified; R3F for hero 3D
****Fonts**** · Fraunces (display), Manrope (body)
****Build**** · `next build` must succeed; case slugs statically generated
****Hosting**** · Static-friendly deploy on Vercel from GitHub
****Privacy**** · No required tracking cookies for core UX
****Browser**** · Modern Chromium/Firefox/Safari; graceful degradation without WebGL

10. Motion requirements (product-level)

Area · Library · Required behavior

Work cards · Framer + GSAP · Enter stagger; hover 3D tilt on fine pointer
Skills cards · Framer + GSAP · Enter stagger; spring hover lift/scale/dim
Testimonials · GSAP · Enter + idle float; CSS hover lift
Hero text · GSAP · Load timeline + word mask
Hero 3D · R3F + GSAP · Entrance, idle, scroll scrub; off on mobile
Navbar · GSAP + CSS · Shrink on scroll; logo breath/spin
About · GSAP · Desktop active step / mobile reveal
Services · GSAP + CSS · Enter + CSS accordion
FAQ · GSAP + Framer · Enter + height accordion
Contact / Footer · GSAP · Stagger / wordmark / mouse glow
Case study page · CSS only · No motion library required
Architecture SVG · None · Static blueprint

11. Design requirements (summary)

- Dark glass aesthetic; light mode warm paper + navy text.
 - Accents: teal / violet / amber by section & theme.
 - Navbar pill max-width ****800px**** (720 scrolled) — ****same on all pages****.
 - Headings: Fraunces; body: Manrope — ****sizes differ by role**** (not one global H1 size).
 - Full visual token listing: see Design Sheet PDF.
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12. Success metrics / acceptance criteria

Code review / viva — must pass

- [] All homepage sections present in correct order
- [] All three case studies open and render full narrative + architecture
- [] Theme toggle works both ways and persists after reload
- [] Contact WhatsApp / mailto / tel work
- [] Mobile usable (nav, no hero 3D clutter)
- [] Reduced-motion path does not break content
- [] Production build succeeds
- [] Deployed URL loads publicly for reviewer

Quality bar

- [] Case study copy is paragraph-level (not one-liners only)
 - [] Images load (WebP under `public/project-art`)
 - [] No “Ask for a demo” on inner pages
 - [] Design Sheet / docs available for review if requested
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13. Known gaps & follow-ups (honest backlog)

Item · Status

`site.github` still placeholder `https://github.com/` · Gap — should be real profile/repo
Per-page SEO / Open Graph / sitemap · Not in v1
Architecture diagram not light-theme adaptive · Accepted exception
Unused `GlassShards`, `marqueeItems` · Cleanup backlog
GitHub repo visibility may be private · Use Vercel public URL for professors
Server-side contact email · Out of scope v1

14. Delivery & documentation deliverables

Deliverable · Purpose

Live Vercel site · Primary link for professors

GitHub repo · Source (public when owner enables)

Design Sheet PDF · Fonts, sizes, glows, **animations per section**

This PRD · Product requirements & acceptance

`docs/PORTFOLIO-DOCUMENTATION.md` · Living motion/design notes

15. Risks

1. Missing image assets in deploy !' broken work/services visuals.
 2. Heavy 3D on low-end desktop !' must pause when off-screen / respect reduced motion.
 3. Private GitHub !' reviewers blocked if only repo link is shared (mitigate: share Vercel).
 4. Placeholder GitHub URL !' broken social trust.
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16. Sign-off

Role · Name · Date

Product owner · Ufaq Khalid · _____

Reviewer (prof / mentor) · _____ · _____

Definition of done: Acceptance checklist in §12 completed; PRD + Design Sheet available; live URL verified.